

Middle East (UAE, Saudi Arabia, Kuwait) GSM Attribute Balancing of Taxi Segment

QCD - Vehicle Performance and Benchmarking

July 15th, 2026





CONTEXT OF THE REGION



Weather/Climate/Temperature Conditions (Middle East)

Environmental Conditions Relevant for EV Taxis

Conditions	UAE	Saudi Arabia	Kuwait
Climate Profile	Hot desert climate with prolonged summer heat	Extreme desert climate with large seasonal temperature variation	Hot desert climate with some of the highest summer temperatures in GCC
Peak Summer Temperature	Frequently 40-45°C+ (May-Sep)	Frequently 45-50°C+ (May-Sep)	Frequently 45-50°C+ (May-Sep)
Heat Soak Exposure	High due to outdoor parking and urban operation	Very high due to long-duration sun exposure and highway travel	Very high due to limited shade and prolonged vehicle parking
Humidity	Moderate to high in coastal Dubai/Abu Dhabi areas	Generally low inland, higher near coastal regions	Moderate coastal humidity
Sand / Dust Exposure	Moderate	High	High
Rain / Flood Exposure	Low annual rainfall, occasional flash flooding events	Very low rainfall	Very low rainfall
Battery Thermal Stress	High during urban taxi operation	Very High due to heat and long-distance operation	Very High due to extreme summer temperatures
HVAC Demand	Very High cooling demand	Very High cooling demand	Very High cooling demand
Charging Environment	Mature charging ecosystem expanding rapidly	National charging rollout underway	Emerging charging ecosystem
Urban Operating Pattern	Airport transfer, limousine, ride-hailing, tourism fleets	Taxi, airport, commuter, corporate fleets	Urban taxi and government fleet operation
Trip Characteristics	Frequent stop-start urban driving	Mixed urban + long highway trips	Primarily short-to-medium urban trips
Vehicle Usage Intensity	High utilization fleet environment	High utilization fleet environment	High utilization fleet environment

Impact on Market

- Strong government-led EV adoption programs are accelerating market growth, particularly in Saudi Arabia's Vision 2030 and the UAE's transportation electrification initiatives. Fleet operators increasingly view EVs as a viable option for airport transfer, limousine, ride-hailing, and corporate mobility applications.
- Customers prioritize thermal robustness, reliability, and operating uptime over maximum vehicle performance. EV acceptance is strongly linked to confidence in hot-weather operation and charging performance.
- Advanced cabin cooling (A/C) and battery thermal management systems are critical due to prolonged summer temperatures frequently exceeding 45°C, especially in Saudi Arabia and Kuwait.
- Premium mobility users expect fast cabin cooldown after hot-soak parking, strong rear-seat cooling performance, and stable battery charging behavior during extreme heat.

Impact on Battery & Electrical System

Factor	Impact
Extreme ambient temperatures (>45°C)	Higher battery temperature exposure increases thermal stress and long-term degradation risk
Hot-soak parking conditions	Elevated battery and cabin temperatures require aggressive cooling during vehicle startup
Frequent DC fast charging in hot climates	Potential charging derating and accelerated battery aging if thermal management is inadequate
Long-distance highway operation	Sustained battery and inverter cooling demand under high-speed driving
Dust and sand exposure	Increased stress on cooling systems, filters, seals, connectors, and external electrical components
Premium comfort expectation	Greater demand on HVAC, seat ventilation, infotainment, and auxiliary electrical systems

ROAD CONDITIONS AND THEIR IMPACT ON FLEET

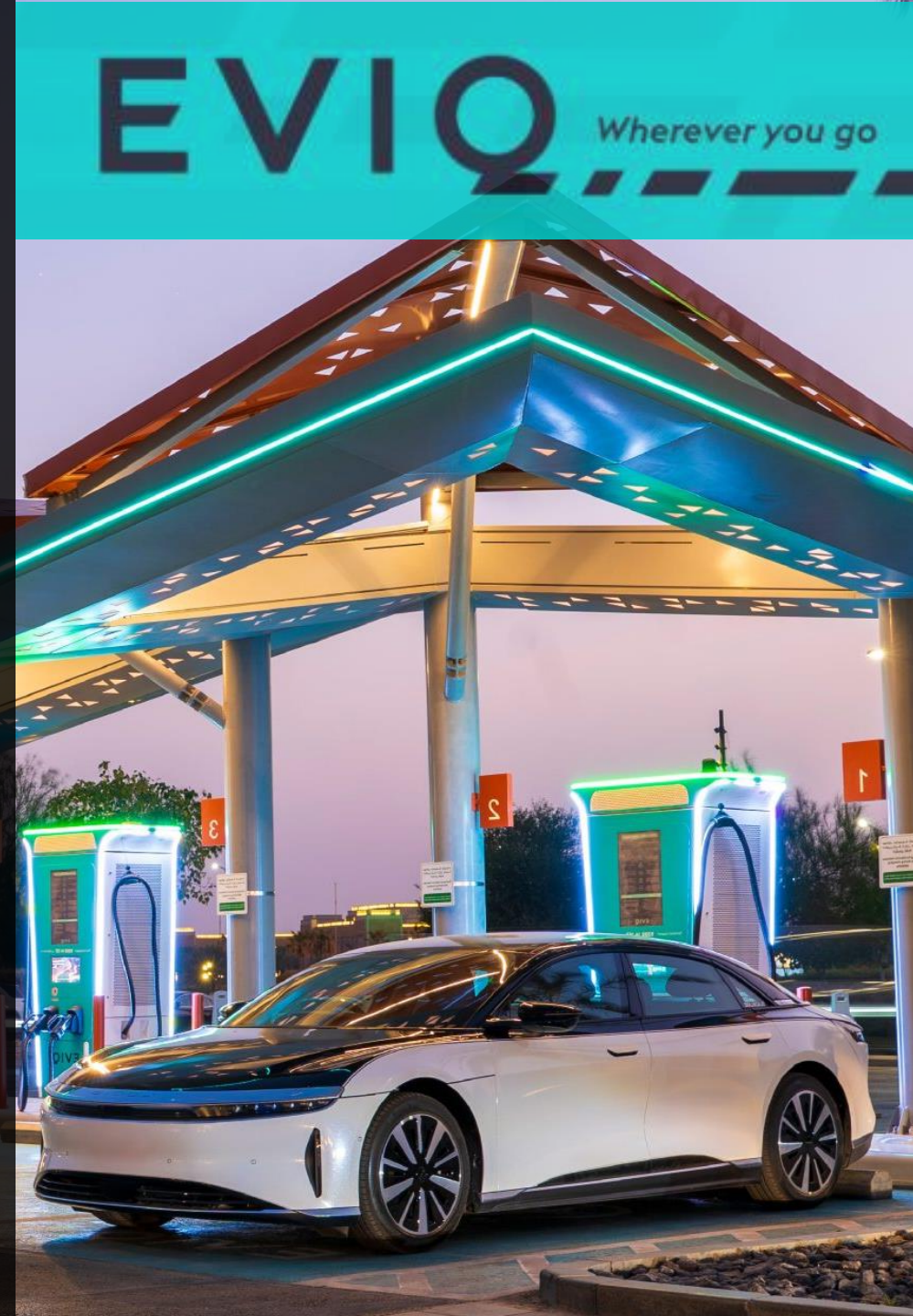
Evaluation Criteria	UAE	Saudi Arabia	Kuwait
Terrain & Traffic Characteristics	Dense urban operation concentrated in Dubai and Abu Dhabi. Heavy congestion during peak periods. Large airport-transfer and limousine market.	Combination of dense urban traffic and long-distance intercity highway travel. Riyadh, Jeddah and Dammam generate significant fleet mileage.	Predominantly urban operation with shorter average trip lengths. City traffic and government fleet operations dominate usage.
Road Quality	Excellent road quality. Modern highways, smooth pavement, and advanced traffic infrastructure.	Excellent highway network. Long-distance expressways create sustained high-speed operating conditions.	Good road infrastructure with modern urban and highway networks. Road quality generally not a major constraint for EV deployment.
Environmental Factors	Extreme summer heat, UV exposure, airborne dust, and coastal humidity.	Extreme desert temperatures, solar loading, dust, and sand exposure.	Extreme summer temperatures and frequent dust exposure. Coastal humidity present in some areas.
Challenges for EVs	Continuous HVAC operation reduces effective range. Hot-soak conditions after outdoor parking.	Long-distance operation under >45°C ambient temperatures creates high thermal load on battery and cooling systems.	Battery cooling and cabin cooling demand remain high throughout the summer season.
Technical & Hardware Solutions	- Enhanced HVAC performance	- Robust battery thermal management	- High-capacity HVAC system
	- Strong battery cooling capability	- High-temperature DC charging calibration	- Optimized thermal insulation
	- UV-resistant interior materials	- Heat-resistant electronics	- Enhanced filtration systems
	- Corrosion-resistant connectors for coastal operation	- Dust-resistant cooling architecture	- Hot-weather battery protection strategy
Operational & Tech Solutions	- Fleet thermal pre-conditioning	- Thermal-aware route planning	- Fleet thermal diagnostics
	- Smart charging scheduling	- Battery temperature monitoring	- Energy optimization for A/C usage
	- Remote cabin cooling before passenger pickup	- Highway charging optimization strategy	- Charging scheduling during lower temperature periods

Driving Behavior - Risks

Driving Behavior Aspect	UAE	Saudi Arabia	Kuwait
Speeding	<u>Frequently cited as a major contributor to crashes. Dubai Police and UAE authorities identify speeding as one of the key risky driving behaviors on highways and urban roads.</u>	<u>Speeding is consistently recognized as a major road safety issue and is commonly associated with aggressive driving and crash involvement. [preprints.org]</u>	One of the most dominant unsafe driving behaviors. A telematics study found overspeeding accounted for 84.89% of detected unsafe driving events.
Distracted Driving / Mobile Phone Usage	<u>Mobile phone distraction is recognized as a significant concern, with specific legislation and enforcement aimed at reducing distracted driving.</u>	Extremely prevalent. A 2024 study found 98% of surveyed drivers reported using mobile phones while driving. Another study found 76.3% admitted texting while driving.	Less documented than in Saudi Arabia, but distraction is recognized as a contributing factor to risky driving and accidents.
Sudden Lane Changes / Lane Swerving	Frequently identified as the leading cause of crashes. Sudden lane deviation caused nearly 2,000 accidents in Dubai over two years.	<u>Aggressive lane changing and overtaking are commonly reported behaviors on major roads and highways. [preprints.org]</u>	<u>Lane cutting and improper lane usage are commonly reported traffic violations. [paaet.edu.kw]</u>
Traffic Rule Violations	Generally lower than neighboring markets due to extensive enforcement systems, but speeding and lane discipline remain concerns.	Violations often involve speeding and mobile phone use while driving.	<u>Frequent violations reported, including queue jumping, illegal lane use, improper U-turns and intersection blockage. [paaet.edu.kw]</u>
Representative Passenger-Car Usage Pattern	High-speed motorway commuting, frequent inter-emirate travel, heavy use of multilane expressways.	Long-distance highway driving, high annual mileage, substantial exposure to high-speed roads and driver distraction risks.	Urban and suburban driving with notable congestion-related aggressive maneuvers and speeding behavior.

Vehicle Charging

	UAE	Saudi Arabia (Riyadh)	Kuwait
Grid Power	230V / 50Hz	230V / 60Hz	240V / 50Hz
AC Standard	Type 2	Type 2	Type 2
DC Standard	CCS2 (Primary)	CCS2 (Primary)	CCS2 (Primary)
Typical AC Power	7.4 / 11 / 22 kW	7.4 / 11 / 22 kW	7.4 / 11 / 22 kW
Typical DC Power	50-150 kW	50-350+ kW	50-150 kW
Public AC Network	Dubai: 4.5k-7k	Riyadh: 0.8k-1.5k	Kuwait City: 0.5k-1k
	Abu Dhabi: 2k-3.5k		
Public DC Network	Dubai: 800-1,200	150-300	80-180
	Abu Dhabi: 300-600		
Charging Maturity	High	Medium-High	Medium
Key Operators	DEWA, ADDC, Tesla	EVIQ, Electromin, Tesla	Kuwait EV Network, ABB
Key Takeaway	Most mature GCC EV charging ecosystem	Fastest charging network growth in GCC	Smaller but growing CCS2 ecosystem



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SEGMENT OVERVIEW AND COMPETITORS



TAXI AND RIDE-HAILING MARKET OVERVIEW



Criteria	UAE	Saudi Arabia	Kuwait
Estimated Population & Fleet Size	UAE taxi market concentrated in Dubai and Abu Dhabi. Large professionally managed fleets operated by RTA franchise companies and airport operators.	Riyadh is one of the largest taxi and ride-hailing markets in the Middle East with rapidly growing mobility demand.	Taxi market centered around Kuwait City with growing demand for app-based transport and fleet services.
Taxi Classification & Operating Zones	Highly regulated taxi system. Licensed operators only. Airport, city, ride-hailing and limousine segments operate under separate permits.	Licensed taxi services coexist with large ride-hailing operations including Uber and Careem.	Regulated taxi operation with airport, city and private hire services.
Usage Rate	High daily taxi utilization driven by tourism, business travel, airport transfers and limited walkability in urban areas.	High utilization due to urban expansion, commuter traffic and increasing demand for on-demand transportation.	High urban utilization, particularly around commercial districts, shopping centers and airport corridors.
Fuel Type & Emission Strategy	Strong government EV adoption strategy. Rapid electrification of taxi and commercial fleets. Charging infrastructure already mature.	Government actively promotes EV adoption through Vision 2030 initiatives. EVIQ leading national charging rollout.	Early-stage EV market. Government and private sector gradually expanding charging infrastructure.
Digital Payment & E-Hailing Rules	Fully digital market. Uber, Careem and operator apps widely used. Cashless payments broadly accepted.	Ride-hailing is mainstream. Mobile payment and digital dispatch are standard customer expectations.	Increasing adoption of ride-hailing and cashless payment solutions.
Mandatory Driver Identity & Vehicle Rules	Commercial operators require licensed drivers and approved vehicles. Fleet age and vehicle inspection requirements strictly enforced.	Licensed commercial operation required. Vehicle compliance and localization requirements managed through transport authorities.	Licensed commercial drivers and approved vehicle registration required.
Taxi Inspection Requirements	Periodic technical inspections required. Fleet uptime, safety compliance and operator service quality heavily monitored.	Regular vehicle inspections and commercial licensing compliance required.	Periodic inspections required to maintain commercial operating permits.

Taxi Electric Vehicle Portfolio In Middle East



Feature	Tesla Model 3	BYD Seal	BYD Atto 3	Hyundai Ioniq 5
Segment	D Sedan	D Sedan	C SUV	C SUV
Number of Seats	5	5	5	5
Powertrain	RWD	RWD	FWD	RWD
Battery Type	Li-ion	LFP Blade	LFP Blade	Li-ion
Battery Voltage	400V	800V	400V	800V
Energy (kWh)	57.5	82.5	60.5	84
Charging Time (DC)	10-80% ~25 min	10-80% ~26 min	10-80% ~30 min	10-80% ~18 min
Motor Power	283 hp	308 hp	201 hp	225 hp
Top Speed	201 km/h	180 km/h	160 km/h	185 km/h
Electric Range WLTP	513 km	570 km	420 km	570 km
MSRP (USD)	~\$42k	~\$41k	~\$25k	~\$48k

Taxi ICE Vehicle Portfolio In Middle East



Feature	Toyota Camry	Hyundai Sonata	Nissan Altima	Toyota Innova
Segment	D-Sedan	D-Sedan	D-Sedan	MPV
Number of Seats	5	5	5	7
Powertrain	FWD	FWD	FWD	RWD
Engine Power	201 hp	191 hp	188 hp	172 hp
Engine Torque	240 Nm	245 Nm	244 Nm	245 Nm
MSRP (USD)	~\$28k	~\$29k	~\$27k	~\$31k
Key Driver Benefits	Reliability, comfort, resale value	Comfort, spacious cabin, fleet friendly	Ride comfort, fuel efficiency	High passenger capacity, durability
Market Status	Taxi Benchmark Leader	Common Taxi & Fleet Vehicle	Major Ride-Hailing Vehicle	Airport / Family Taxi
Fuel Consumption (Combined)	6.0-6.8 L/100km	6.7-7.5 L/100km	6.5-7.0 L/100km	8.5-9.5 L/100km
Estimated Weekly Fuel Cost*	\$40-50	\$45-55	\$40-50	\$55-70



ATTRIBUTE PRIORITY FOR THE MARKET OF MIDDLE EAST





THERMAL ATTRIBUTE: HIGH



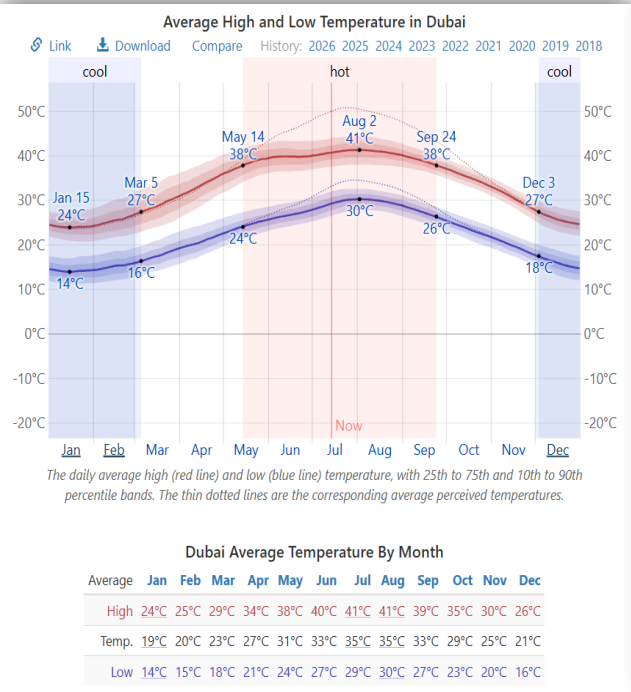


THERMAL ATTRIBUTE

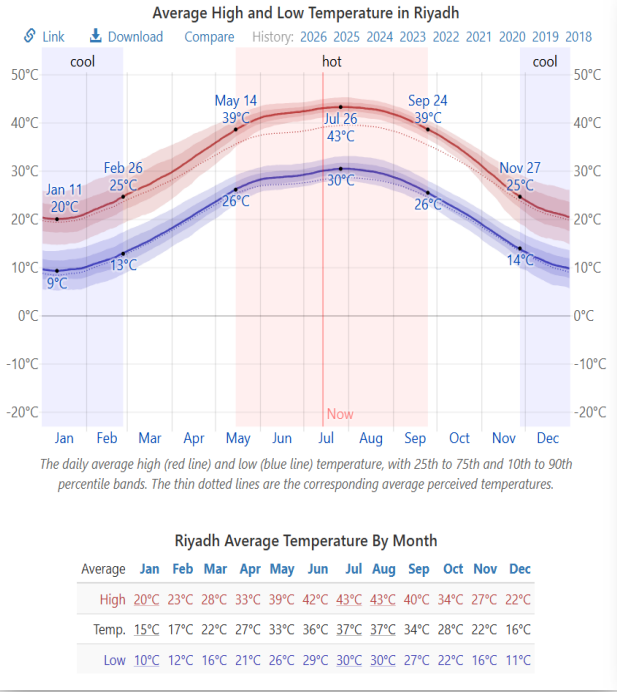
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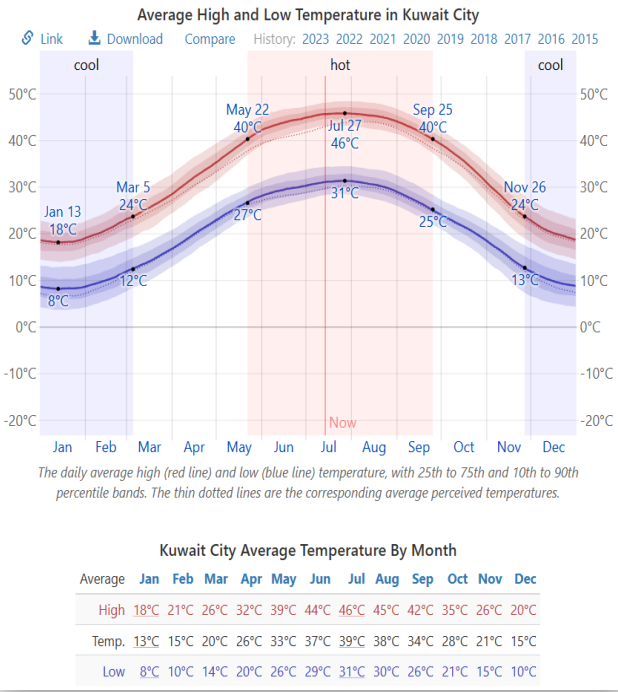
1. Ambient Temperature	UAE - Dubai	Saudi Arabia - Riyadh	Kuwait City
T-AMBIENT annual average	27°C	26°C	26°C
T-HOT annual maximum (peak summer daily max)	42°C	43°C	46°C
T-COLD annual minimum (peak winter daily min)	14°C	9°C	8°C
Number of days >35°C	Up to 190 days (Apr – Oct)		
Number of days >40°C	Up to 120 days (May – Sept)		
Number of days >45°C	None	1-2 days	Up to 60 days (Jul – Aug)
Number of days >50°C	None	None	1-2 days



DUBAI: all-time average High-Low Temperature



RIYADH: all-time average High-Low Temperature



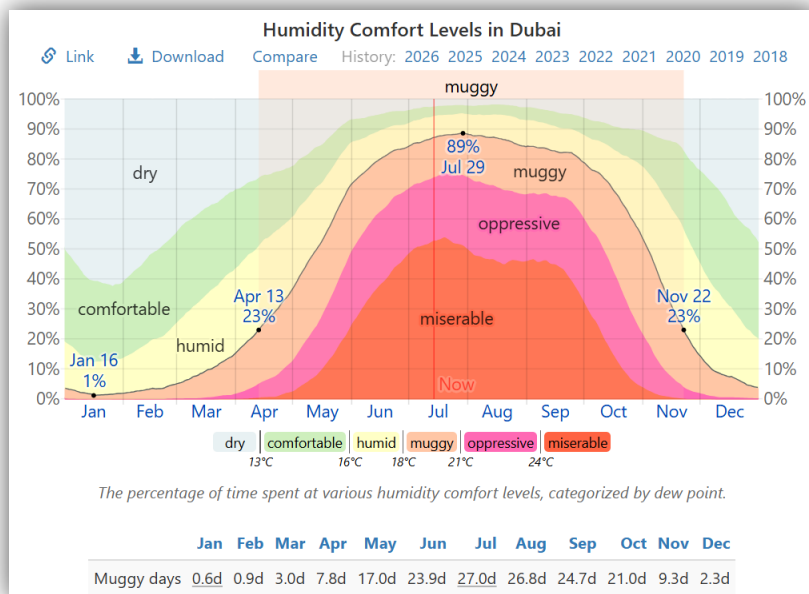
KUWAIT CITY: all-time average High-Low Temperature



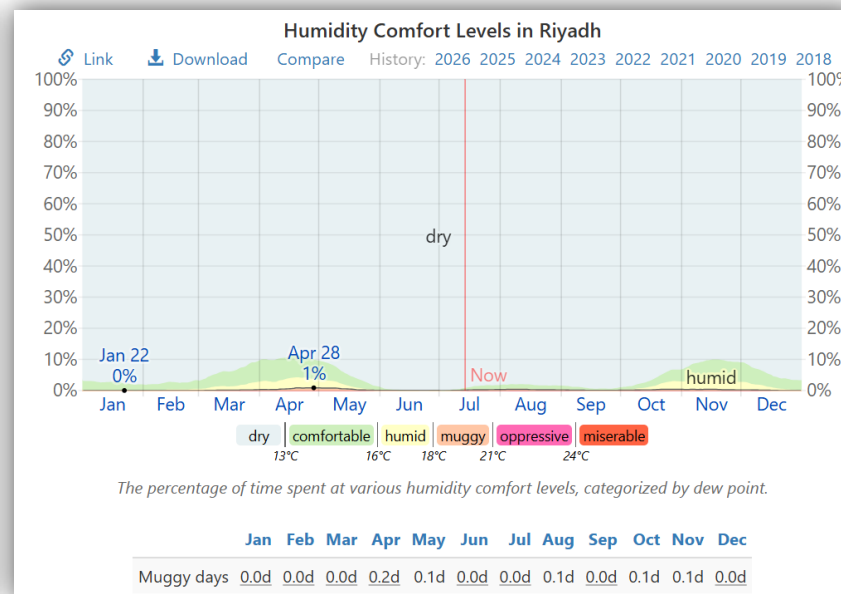


2. Humidity	UAE - Dubai	Saudi Arabia - Riyadh	Kuwait City
HUMIDITY annual average	59-69% RH*	23-32% RH	29-33% RH

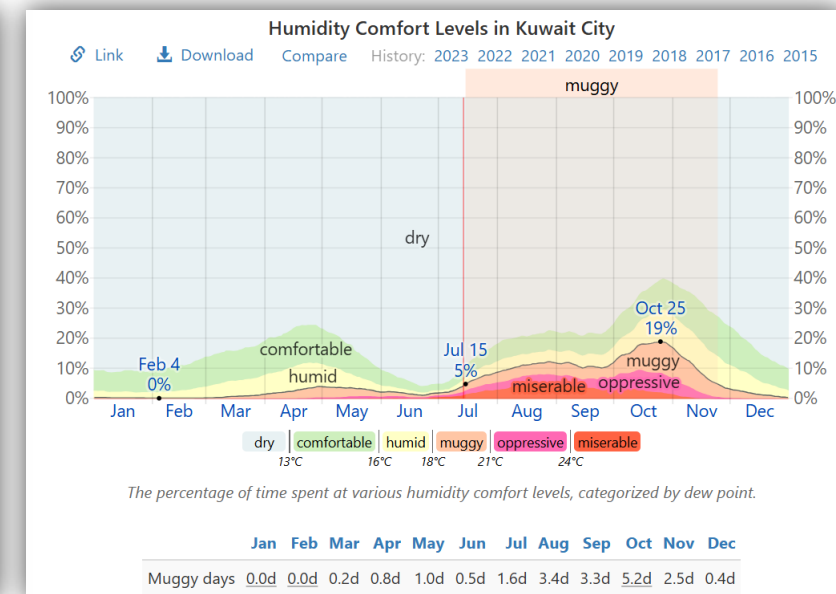
*RH = Relative Humidity



DUBAI: all-time average humidity



RIYADH: all-time average humidity



KUWAIT CITY: all-time average humidity



3. Solar Load Severity

Parameter	UAE - Dubai	Saudi Arabia - Riyadh	Kuwait City
Solar radiation intensity (GHI)	6.05-6.12 kWh/m ² /day	6.21 kWh/m ² /day	4.93 kWh/m ² /day
Peak solar radiation	1041-1400 W/m ²	1115 W/m ²	1050 W/m ²
UV index	11-12+	11-12+	11-12+



4. Gradeability & Terrain

Parameter	UAE - Dubai	Saudi Arabia - Riyadh	Kuwait City
MAX Elevation Above Sea Level	1,042 m	835 m	98 m
MIN Elevation Above Sea Level	-9 m	535 m	-3 m
AVG Elevation Above Sea Level	64 m	630 m	10 m
Major Incline Locations	Parking ramps, flyovers, highway interchanges	Wadi Hanifa escarpments, western districts, highway grades	Parking ramps, flyovers, highway interchanges





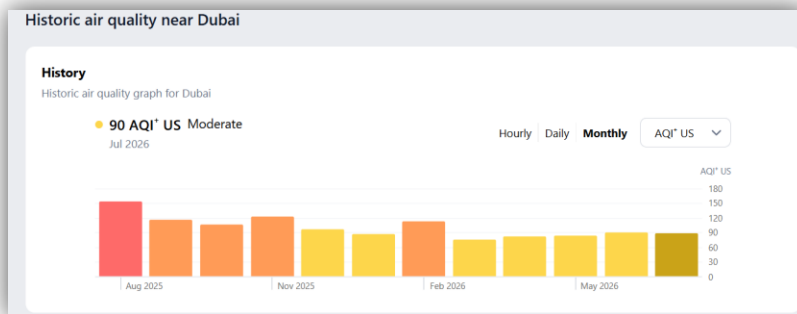
5. High-Speed Operation Severity

Parameter	UAE - Dubai	Saudi Arabia - Riyadh	Kuwait City
Average highway speed	115-130 km/h	110 km/h	105 km/h
Maximum legal speed	120-140 km/h	120 km/h	120 km/h
Sustained high-speed duration	30-90 min	30-45 min	20-40 min
Desert-highway operation ratio	40-60%	70%	85%

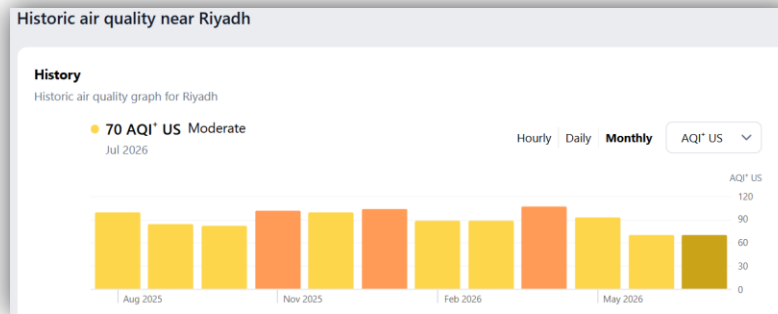


6. Environmental Exposure Severity

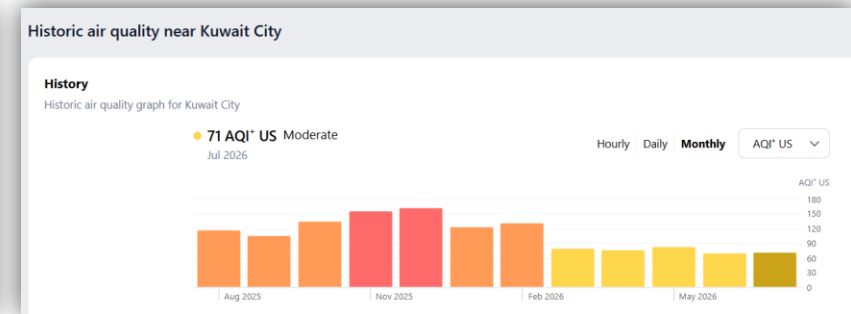
Parameter	UAE - Dubai	Saudi Arabia - Riyadh	Kuwait City
Sandstorm frequency	10-12 events/year	~28 events/year	~35 events/year
Coastal operation ratio	85-90%	0%	75%
Flood exposure	<3-5 days/year	<2 days/year	<3 days/year
Primary Pollutant	PM2.5, dust	PM2.5, desert dust	PM10, PM2.5, desert dust



DUBAI: AQI Index



RIYADH: AQI Index



KUWAIT CITY: AQI Index





VEH. DYNAMICS ATTRIBUTE: HIGH



UAE (Abu Dhabi / Dubai)

- Broad, straight highways with 6-8 lanes per direction dominate the taxi operating environment.
- Roundabout density is low and primarily limited to suburban communities and older districts.
- Highway interchange density is extremely high, featuring complex multi-level cloverleafs and flyovers.
- Curve geometry is dominated by long-radius sweepers and gentle arcs optimized for high-speed travel.
- Lane-change frequency is high due to multi-lane expressways and high-speed traffic flow.
- Highway merge frequency is high as vehicles transition between radial and axial expressways.
- Obstacle avoidance frequency is low because of proactive road maintenance and excellent surface quality.
- Emergency maneuvers occur primarily during sudden high-speed braking events.
- Parking maneuver frequency is high around airports, shopping malls, hotels, and passenger pickup areas.
- Vehicle dynamics tuning should prioritize high-speed stability, crosswind robustness, and lane-change confidence.

Saudi Arabia (Riyadh)

- Extensive urban grid network combined with modern ring roads creates a demanding maneuvering environment.
- Roundabout density is high throughout the city road network.
- Driver non-compliance at roundabouts is common, increasing dynamic maneuvering demand.
- Curve-radius distribution is predominantly tight, requiring frequent low-speed steering corrections.
- Lane-change frequency is extremely high, particularly in congested urban corridors.
- Obstacle avoidance frequency is high due to aggressive traffic behavior and dense traffic conditions.
- Highway merge zones are frequently congested and require rapid driver response.
- Emergency maneuver probability is high because of abrupt cut-ins, lane violations, and roundabout conflicts.
- Parking maneuvers are extremely frequent due to roadside pickup/drop-off operations and unmanaged parking.
- Vehicle dynamics tuning should emphasize steering responsiveness, agility, brake confidence, and transient stability.

Kuwait City

- Concentric ring-road architecture creates a highly structured but demanding driving environment.
- Highway interchange density is high with numerous closely spaced junctions.
- Curve-radius distribution is mixed, with many tight-radius ramps and loop exits.
- Lane-change frequency is high on multi-lane ring roads and expressways.
- Obstacle avoidance frequency is high due to potholes, pavement rutting, loose aggregate, and surface defects.
- Highway merges occur frequently and often require rapid speed adjustments.
- Emergency maneuver probability is high because of sudden braking caused by road anomalies and surface hazards.
- Parking maneuver frequency is high in congested urban areas with limited parking space.
- Vehicle dynamics tuning should prioritize ride isolation, suspension robustness, pothole compliance, and directional stability.
- Steering and chassis systems must balance maneuverability with durability under rough-road conditions.





UAE (Abu Dhabi / Dubai)

- Road quality is among the highest globally, with premium pavement standards across taxi operating routes.
- Expressway IRI typically ranges from **0.6-1.2 m/km**, supporting excellent ride comfort and vehicle stability.
- New highways are designed to achieve IRI values below 0.9 m/km on roads above 100 km/h.
- Asphalt roads account for approximately 97% of the network.
- Concrete road exposure is limited to specialized infrastructure such as toll plazas, industrial facilities, and bus lanes.
- Patchwork road frequency is minimal due to rapid repair and resurfacing programs.
- Speed bumps are common in residential communities, school zones, and traffic-calming areas.
- Cobblestone and block pavements are limited to heritage districts and pedestrian-oriented zones.
- Village road exposure is virtually nonexistent for taxi fleets.
- Poor road exposure is below 1% of operating mileage.

Saudi Arabia (Riyadh)

- Road conditions reflect a transition from aging municipal roads to modern rehabilitated corridors.
- Expressway IRI typically ranges from 1.5-2.5 m/km.
- Arterial and local roads exhibit roughness levels between 2.8-4.5 m/km.
- Asphalt roads account for more than 98% of the network.
- Common pavement defects include cracking, raveling, patching, potholes, and local depressions.
- Pothole occurrence is generally concentrated on secondary and residential roads.
- Patchwork road frequency is high due to metro construction and utility expansion activities.
- **Speed bump** density is high near schools, mosques, and community zones.
- Village road exposure remains low but exists in expanding urban fringe developments.
- Poor road exposure accounts for approximately 10-15% of taxi operations in affected areas.
- A large-scale SAR 4 billion rehabilitation program is improving network quality and ride smoothness.

Kuwait City

- Kuwait presents the most challenging pavement environment among the three markets.
- Expressway IRI typically ranges from 3.5-6.5 m/km, substantially rougher than UAE and Saudi Arabia.
- Local road IRI can reach critical levels above 40 m/km in severely deteriorated sections.
- Heavy rainfall events since 2018 accelerated pothole formation and moisture-induced pavement damage.
- Extreme summer temperatures above 50°C contribute to rapid asphalt aging and binder degradation.
- Asphalt roads account for more than 99% of total road infrastructure.
- Patchwork road frequency is extremely high due to extensive maintenance and repair activity.
- Pothole frequency has historically been extremely high across many urban corridors.
- Speed bumps are common and often have irregular geometries.





Gradeability & Terrain

Parameter	UAE - Dubai	Saudi Arabia - Riyadh	Kuwait City
MAX Elevation Above Sea Level	1,042 m	835 m	98 m
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Driving Behavior - Risks

Driving Behavior Aspect	UAE	Saudi Arabia	Kuwait
Speeding	<u>Frequently cited as a major contributor to crashes. Dubai Police and UAE authorities identify speeding as one of the key risky driving behaviors on highways and urban roads.</u>	<u>Speeding is consistently recognized as a major road safety issue and is commonly associated with aggressive driving and crash involvement.</u> [preprints.org]	One of the most dominant unsafe driving behaviors. A telematics study found overspeeding accounted for 84.89% of detected unsafe driving events.
Distracted Driving / Mobile Phone Usage	<u>Mobile phone distraction is recognized as a significant concern, with specific legislation and enforcement aimed at reducing distracted driving.</u>	Extremely prevalent. A 2024 study found 98% of surveyed drivers reported using mobile phones while driving. Another study found 76.3% admitted texting while driving.	Less documented than in Saudi Arabia, but distraction is recognized as a contributing factor to risky driving and accidents.
Sudden Lane Changes / Lane Swerving	Frequently identified as the leading cause of crashes. Sudden lane deviation caused nearly 2,000 accidents in Dubai over two years.	<u>Aggressive lane changing and overtaking are commonly reported behaviors on major roads and highways.</u> [preprints.org]	<u>Lane cutting and improper lane usage are commonly reported traffic violations.</u> [paaet.edu.kw]
Traffic Rule Violations	Generally lower than neighboring markets due to extensive enforcement systems, but speeding and lane discipline remain concerns.	Violations often involve speeding and mobile phone use while driving.	<u>Frequent violations reported, including queue jumping, illegal lane use, improper U-turns and intersection blockage.</u> [paaet.edu.kw]
Representative Passenger-Car Usage Pattern	High-speed motorway commuting, frequent inter-emirate travel, heavy use of multilane expressways.	Long-distance highway driving, high annual mileage, substantial exposure to high-speed roads and driver distraction risks.	Urban and suburban driving with notable congestion-related aggressive maneuvers and speeding behavior.



EMDQ ATTRIBUTE: HIGH





Parameter	UAE (Dubai/Abu Dhabi)	Saudi Arabia (Riyadh)	Kuwait (Kuwait City)
Electricity Cost (Public DC Charging)	USD 0.32/kWh	USD 0.32-0.66/kWh	USD 0.25/kWh
Electricity Cost (Residential/Home Charging)	USD 0.07-0.10/kWh	USD 0.16/kWh	USD 0.007/kWh
Electricity Cost (Commercial Charging)	USD 0.054/kWh	N/A	N/A
Fuel Cost Benchmark (95 Octane)	USD 0.90/L	USD 0.62/L	USD 0.34/L





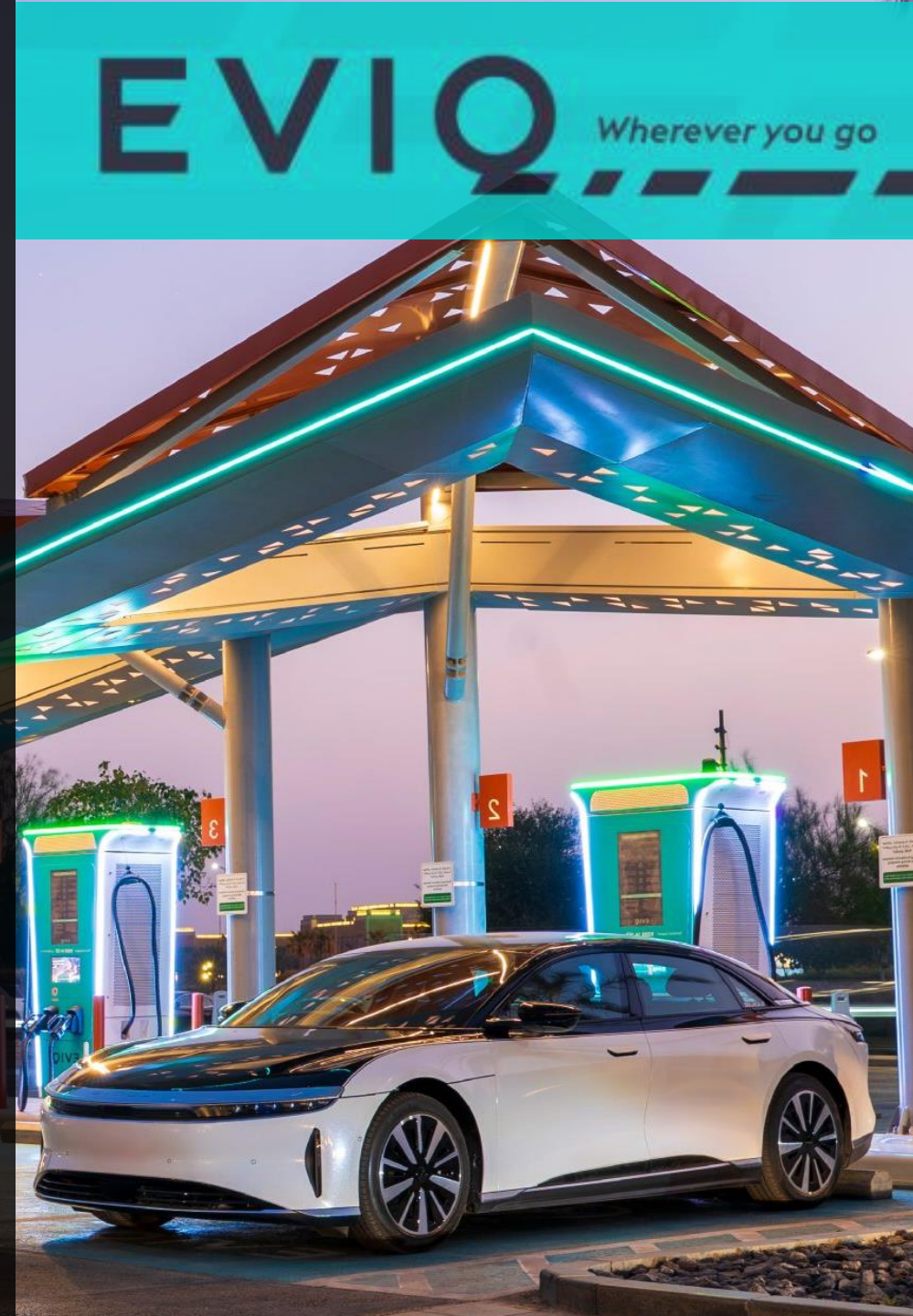
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DURABILITY ATTRIBUTE: HIGH





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- Patchwork road frequency is minimal due to rapid repair and resurfacing programs.
- Speed bumps are common in residential communities, school zones, and traffic-calming areas.
- Cobblestone and block pavements are limited to heritage districts and pedestrian-oriented zones.
- Village road exposure is virtually nonexistent for taxi fleets.
- Poor road exposure is below 1% of operating mileage.

Saudi Arabia (Riyadh)

- Road conditions reflect a transition from aging municipal roads to modern rehabilitated corridors.
- Expressway IRI typically ranges from 1.5-2.5 m/km.
- Arterial and local roads exhibit roughness levels between 2.8-4.5 m/km.
- Asphalt roads account for more than 98% of the network.
- Common pavement defects include cracking, raveling, patching, potholes, and local depressions.
- Pothole occurrence is generally concentrated on secondary and residential roads.
- Patchwork road frequency is high due to metro construction and utility expansion activities.
- **Speed bump** density is high near schools, mosques, and community zones.
- Village road exposure remains low but exists in expanding urban fringe developments.
- Poor road exposure accounts for approximately 10-15% of taxi operations in affected areas.
- A large-scale SAR 4 billion rehabilitation program is improving network quality and ride smoothness.

Kuwait City

- Kuwait presents the most challenging pavement environment among the three markets.
- Expressway IRI typically ranges from 3.5-6.5 m/km, substantially rougher than UAE and Saudi Arabia.
- Local road IRI can reach critical levels above 40 m/km in severely deteriorated sections.
- Heavy rainfall events since 2018 accelerated pothole formation and moisture-induced pavement damage.
- Extreme summer temperatures above 50°C contribute to rapid asphalt aging and binder degradation.
- Asphalt roads account for more than 99% of total road infrastructure.
- Patchwork road frequency is extremely high due to extensive maintenance and repair activity.
- Pothole frequency has historically been extremely high across many urban corridors.
- Speed bumps are common and often have irregular geometries.





VEHICLE HANDLING FREQUENCIES

SOURCE

(contact [anhbv12 / tienvx2](#) for access)

UAE (Dubai & Abu Dhabi)

- Passenger turnover is exceptionally high due to continuous 24-hour fleet operation.
- Typical taxis complete approximately **24-28 passenger trips per day**.
- Daily door opening and closing frequency reaches approximately **48-112 cycles/day**.
- Airport transfer activity generates frequent luggage handling events.
- Tailgate or trunk opening frequency averages **20-30 cycles/day**.
- Hood opening frequency averages approximately **1.5 cycles/day** during inspections and shift handovers.
- Passenger turnover rate is the highest among the three markets.
- EV fleet deployment is accelerating through Dubai Taxi Company electrification programs.
- Charging-port opening frequency averages **3-5 cycles/day** for active EV taxis.
- Charging-port mechanisms must withstand dust, heat, and repeated daily operation.

Saudi Arabia (Riyadh)

- Longer average trip distances reduce daily passenger turnover compared with Dubai.
- Typical taxis complete approximately **12-18 passenger trips per day**.
- Daily door opening and closing frequency ranges from **24-72 cycles/day**.
- Tailgate opening frequency averages approximately **8-15 cycles/day**.
- Most urban trips involve limited luggage handling requirements.
- Hood opening frequency remains low and is generally associated with routine fleet inspections.
- Ride-hailing activity is extremely high through Uber, Careem, Bolt, Jeeny, and other digital platforms.
- Passenger turnover remains high due to strong app-based mobility demand.
- EV adoption is increasing under Vision 2030 electrification targets.
- Charging-port opening frequency averages **3-5 cycles/day** for active EV taxis.
- Charging-port systems must tolerate high ambient temperatures and dust intrusion.

Kuwait (Kuwait City)

- High-density urban taxi service creates moderate-to-high passenger turnover.
- Typical taxis complete approximately **15-22 passenger trips per day**.
- Daily door opening and closing frequency ranges from **30-88 cycles/day**.
- Tailgate opening frequency averages approximately **12-20 cycles/day**.
- Airport-transfer vehicles and family-oriented vans experience elevated cargo-area usage.
- SUV and MPV fleets generate additional tailgate wear due to luggage operations.
- Hood opening frequency remains relatively low and maintenance-driven.
- Ride-hailing penetration is lower than UAE and Saudi Arabia.
- Passenger turnover rates are lower than UAE but higher than traditional low-utilization fleets.
- EV adoption remains limited, resulting in lower charging-port cycling frequencies.
- Future EV deployments are expected to increase charging-port usage substantially.





Common among 3 markets:

- Passenger-inflicted door slamming is frequent due to very high passenger turnover rates.
- Door latches, hinge brackets, door checks, and interior trim experience continuous mechanical loading.
- Tailgate and trunk slamming severity is high due to intensive airport-transfer operations.
- Dedicated airport taxi fleets generate repeated luggage-loading impacts throughout the day.
- Frequent trunk operation increases risk of latch wear, hinge fatigue, seal degradation, and alignment drift.
- Charging-port handling abuse is emerging as EV penetration increases.
- Drivers may force DC charging connectors into ports during rapid turnaround operations.
- Charging-port housings, guide pins, lock actuators, and flap hinges experience elevated wear.
- Ambient temperatures above 45°C accelerate plastic aging and fatigue cracking around charging-port assemblies.
- Dust intrusion further increases wear of charging-port mechanisms and seals.
- Fleet abuse exposure is moderate but continuous due to 24-hour taxi utilization.
- Critical durability items include door systems, trunk hardware, charging ports, weather seals, and lock actuators.





UAE (Dubai & Abu Dhabi)

- Public taxis must be retired after approximately **4 years** for gasoline and hybrid vehicles.
- Fully electric taxis may remain in operation for up to **6 years**.
- Fleet replacement cycles are tightly regulated to maintain service quality and reliability.
- Manufacturer warranty expectation is typically **3 years / 100,000 km**.
- Fleet operators prioritize drivetrain durability over minimum warranty compliance.
- Reliability expectations exceed **500,000 km drivetrain life**.
- Commercial uptime requirements exceed **95% fleet availability**.
- Vehicle downtime directly impacts fleet profitability and service capacity.
- Customers expect high reliability, low downtime, and predictable maintenance costs.

Saudi Arabia (Riyadh)

- Taxi operating life is legally limited to **5 years from the year of manufacture**.
- Fleet replacement cycles are driven by TGA regulatory requirements.
- Manufacturer warranty expectation is typically **3 years / 100,000 km**.
- Vehicles are expected to withstand **more than 500,000 km** during their regulated service life.
- High-speed and high-temperature operation is considered a normal durability requirement.
- Drivetrain longevity is a critical purchasing criterion.
- Commercial uptime requirements generally exceed **90% availability**.
- Residual value sensitivity is very high because retired taxis are commonly exported to secondary markets.
- Customers expect vehicles to complete their full regulatory lifecycle without major failures.

Kuwait (Kuwait City)

- Taxi replacement cycles are more flexible than in UAE and Saudi Arabia.
- Typical fleet replacement occurs every **5-7 years**.
- Vehicle ownership and operating lifecycles are less tightly regulated.
- Manufacturer warranty expectation is typically **3 years / 100,000 km**.
- Reliability expectations exceed **400,000 km drivetrain life**.
- Commercial uptime requirements exceed **90% fleet availability**.
- Low-cost maintenance is a major fleet purchasing criterion.
- Residual value sensitivity remains high within the second-hand vehicle market.
- Highly durable Japanese and Korean sedans dominate fleet purchasing decisions.
- Durability expectations prioritize affordability, uptime, and repair simplicity.





COMPARISON TABLE OF GSM MARKETS



Comparison Table of GSM Markets – Middle East

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[123 - Copy for VPBO](#)



Country	Launch date	Proposed vehicle from VDD	LHD / RHD	Geography / urban context	Climate & temperature condition	Road & infrastructure condition	Driving behavior & customer expectation	Attribute priority for market and risk
UAE - TBC	TBU	Limo Green	LHD	Likely Dubai/Abu Dhabi urban taxi environment with relatively mature EV policy and charging ecosystem.	Dubai climate shows persistent summer heat above 40°C with hot-desert operating conditions.	Road quality and charging deployment are stronger than many neighboring markets; the challenge is meeting a high benchmark, not simply entering the market.	Customers/fleets will expect premium thermal comfort, fast charging turnaround, clean operations, and strong reliability from launch.	HIGH = THM, VD, EMDQ, DUR MEDIUM = CORR, SAF, NVH, UE, EEU, EEP, PQ, A&U, TIGHT, ENVI, AERO, ADAS LOW = WT
Saudi Arabia - Riyadh	22-Oct-27	VF6 Legacy (ME) & Limo Green	LHD	Desert megacity / airport / commuter taxi duty with extreme summer heat.	Hot desert climate; average summer daytime temperatures are around 45°C.	Road infrastructure is improving with EV charging rollout, but the dominant engineering issue is heat, not pavement roughness.	Customers expect very strong A/C, fast cabin cooldown, stable performance in solar soak, and acceptable charging/thermal behavior despite heat.	HIGH = THM, VD, EMDQ, DUR MEDIUM = CORR, SAF, NVH, UE, EEU, EEP, PQ, A&U, TIGHT, ENVI, AERO, ADAS LOW = WT
Kuwait	8-Oct-27	VF6 Legacy (ME)	LHD	Gulf urban fleet market with very high thermal stress and developing EV infrastructure.	Market analysis explicitly highlights extreme climate conditions and the need for advanced EV thermal management.	Roads are not the main problem; heat, charging behavior, and cabin comfort are.	Drivers/fleets will judge the car on hot-weather range stability, strong A/C, and heat-soak recovery rather than on rough-road survival.	HIGH = THM, VD, EMDQ, DUR MEDIUM = CORR, SAF, NVH, UE, EEU, EEP, PQ, A&U, TIGHT, ENVI, AERO, ADAS LOW = WT



Comparison Table of GSM Markets

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Country	Launch date	Proposed vehicle from VDD	LHD / RHD	Geography / urban context	Climate & temperature condition	Road & infrastructure condition	Driving behavior & customer expectation	Attribute priority for market and risk
UK - <i>Liverpool/Manchester/ Birmingham</i>	23-Apr-27	VF6N	RHD	Dense Northern UK urban taxi environment, RHD market, often rough surface exposure.	Temperate and wet; climate is not extreme but consistent rain/cool operation affects comfort and visibility.	The UK pothole problem is severe, with large report volumes and compensation claims; potholes are a direct fleet durability threat.	Drivers expect good ride over broken roads, tight turning, strong steering feel, reliable electronics, and low downtime.	HIGH = DUR, VD, EMDQ, SAF, NVH, ADAS MEDIUM = THM, WT, CORR, UE, EEU, EEP, PQ, A&U, TIGHT, ENVI, AERO LOW = WT
UK - London	17-Sep-27	VF6N	RHD	Global benchmark taxi city, RHD, very dense urban / airport / curbside service environment. 192021	Temperate maritime climate with mild winters and regular rain. 18	Roads are workable but urban operations are intense; the real barrier is regulation and operational fit, not raw road survival. 192022	Customers and regulators expect ZEC compliance, accessibility, refined ingress/egress, strong rear-seat comfort, quietness, and polished ADAS behavior.	HIGH = DUR, VD, EMDQ, SAF, NVH, ADAS MEDIUM = THM, WT, CORR, UE, EEU, EEP, PQ, A&U, TIGHT, ENVI, AERO LOW = WT
Finland - Helsinki	9-Apr-27	VF6 Legacy	LHD	Mature Nordic EV market with high user knowledge and strong winter operating expectations.	Helsinki has cold, humid, snowy winters and cool summers; EV adoption is already very high.	Roads are generally good, but winter contamination / moisture and possible salt exposure raise corrosion and usability demands.	Customers will compare your car against mature EV winter behavior—range prediction, preconditioning, cabin warm-up, and winter charging must be very good.	HIGH = THM, UE, EMDQ, DUR, CORR, TIGHT, SAF MEDIUM =, WT, NVH, UE, EEU, EEP, PQ, A&U, TIGHT, ENVI, AERO, VD, ADAS LOW = NA
Canada	19-Feb-27	VF8 Legacy	LHD	Large urban + airport + suburban taxi duty; cold-season operations dominate launch risk.	Ottawa-type climate is freezing, snowy winter / warm summer; official winter EV testing found 14–39% real-world range loss at -7°C to -15°C.	National road quality is comparatively good, but winter potholes, slush, salt, and curb-edge impacts still matter.	Customers will expect honest winter range, fast cabin warm-up, strong demist/defrost, and dependable cold-soak starting/charging.	HIGH = EMDQ, EEU, EEP, DUR, CORR, TIGHT MEDIUM = THM, ENVI, AERO, ADAS, NVH, VD, PQ, A&U LOW = WT, UE, SAF





Comparison Table of GSM Markets - Europe

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Country	Launch date	Proposed vehicle from VDD	LHD / RHD	Geography / urban context	Climate & temperature condition	Road & infrastructure condition	Driving behavior & customer expectation	Attribute priority for market and risk
France - Lyon/ Bordeaux	14-May-27	VF6NP → VF6 Legacy	LHD	Mature French EV markets; Lyon adds hot summers, Bordeaux adds humid/coastal flavor.	Lyon sees hot summers and meaningful rainfall; Bordeaux is not in the climate source set here, but coastal humidity is a sensible operating assumption for your launch planning.	Infrastructure is mature; this is more a refinement and expectation market than a survival market.	Customers expect good software/HMI, smooth ADAS, comfortable ride, and low cabin noise because the EV market is already broad.	HIGH = SAF MEDIUM = THM, DUR, VD, EMDQ, CORR, NVH, UE, EEU, EEP, PQ, A&U, ENVI, WT, AERO, ADAS LOW = TIGHT
France - Paris	17-Sep-27	VF6N - SOP: 06/2027	LHD	Dense European capital with mature EV usage and strong urban customer expectations.	Temperate climate; no major thermal extreme.	Roads are serviceable; the main issue is dense stop-go urban operation and benchmark expectations.	Drivers/passengers will judge rear-seat comfort, ingress/egress, app/HMI stability, ADAS smoothness, and uptime.	HIGH = SAF, VD MEDIUM = THM, DUR, EMDQ, CORR, NVH, UE, EEU, EEP, PQ, A&U, ENVI, WT, AERO, ADAS LOW = TIGHT
Spain - Madrid	15-Oct-27	VF6N - SOP: 06/2027	LHD	Large inland capital with expanding public charging and strong urban/peri-urban taxi demand.	Continental/Mediterranean pattern with hot dry summers and cool winters.	Roads are generally workable; charging experience and urban destination charging are more important than road harshness.	Drivers will care about HVAC efficiency, charge confidence, and smooth stop-go response in summer city use.	HIGH = SAF MEDIUM = THM, DUR, VD, EMDQ, CORR, NVH, UE, EEU, EEP, PQ, A&U, TIGHT, ENVI, WT, AERO, ADAS LOW = TIGHT



Country	Launch date	Proposed vehicle from VDD	LHD / RHD	Geography / urban context	Climate & temperature condition	Road & infrastructure condition	Driving behavior & customer expectation	Attribute priority for market and risk
Austria - Vienna	21-May-27	VF6 Legacy	LHD	Central European capital, mature EV market, dense city with tram and urban mixed-surface use.	Warm summers, cold winters near freezing, moderate precipitation year-round.	Road quality is strong; the bigger issue is refinement over urban infrastructure rather than damage survival.	Customers will compare against established European EV standards: ADAS smoothness, quality feel, comfortable ride, winter demist, and quiet cabin.	HIGH = SAF MEDIUM = THM, DUR, VD, EMDQ, CORR, NVH, UE, EEU, EEP, PQ, A&U, TIGHT, ENVI, WT, AERO, ADAS LOW = WT
Portugal - Lisbon + Porto	12-Mar-27	VF6 Legacy	LHD	Atlantic coastal urban markets with strong EV familiarity; Lisbon is drier/milder, Porto is wetter.	Lisbon is mild Mediterranean; Porto is cooler/wetter oceanic with much higher annual rain exposure.	Road quality is strong by global comparison; infrastructure is good but users still complain about charging sufficiency / wait time in some cases.	Customers are already used to EVs and interoperable charging, so they notice software, charge stability, rear-seat comfort, and overall polish quickly.	HIGH = MEDIUM = EMDQ, WT, CORR, SAF, NVH, THM, DUR, VD, UE, EEU, EEP, PQ, A&U, TIGHT, ENVI, WT, AERO, ADAS LOW = THM, DUR, VD



Comparison Table of GSM Markets - Africa

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Country	Launch date	Proposed vehicle from VDD	LHD / RHD	Geography / urban context	Climate & temperature condition	Road & infrastructure condition	Driving behavior & customer expectation	Attribute priority for market and risk
Ghana - Accra	22-Oct-27	Heriogreen Legacy	LHD	Coastal tropical capital with recurring urban flooding and developing EV ecosystem.	Hot, humid, rainy-season exposure; flood events are recurrent and operationally serious.	Roads can become impassable during heavy rain; hidden potholes, standing water, underbody strikes, and drainage problems are major fleet risks.	Drive behavior is flood-avoidance / stop-go / variable-surface; reliability issues quickly become brand issues because infrastructure is still maturing.	HIGH = THM, DUR, VD, EMDQ, TIGHT, CORR, ADAS MEDIUM = SAF, NVH, UE, EEU, EEP, PQ, A&U, ENVI, WT, AERO LOW =
Ivory Coast - Approved	15-Dec-26	Heriogreen Legacy - SOP 17/08/2026	LHD	Abidjan-type launch should be treated as a tropical coastal urban market with recurrent flood / resilience challenges.	Hot, humid, flood-prone operating pattern; risk is driven by water exposure and urban drainage weakness.	Roads and drainage can both be weak; flood-related damage to roads, services, and urban operations is a known problem.	Customers/fleets need a car that survives wet urban use, stays serviceable, and does not create large unplanned repair costs in an early EV market.	HIGH =DUR, A&U MEDIUM = UE, EEU, EEP, ENVI, WT, AERO, EMDQ, TIGHT, CORR, ADAS LOW = NVH, VD, SAF, THM, PQ



Comparison Table of GSM Markets – Caucasus

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Country	Launch date	Proposed vehicle from VDD	LHD / RHD	Geography / urban context	Climate & temperature condition	Road & infrastructure condition	Driving behavior & customer expectation	Attribute priority for market and risk
Azerbaijan	19-Mar-27	VF6 Legacy	LHD	Baku-type urban launch: dense city traffic with infrastructure modernization underway.	Seasonal climate stress exists but the bigger issue in the sources is operating intensity and infrastructure transition rather than temperature extremes.	Road quality is mid-pack globally; Baku is investing in major transport and road-safety improvements, implying uneven quality outside improved corridors.	Expect urban stop-go, mixed surfaces, and customer sensitivity to down-time / parts delays in a still-developing EV taxi environment.	HIGH = ADAS MEDIUM = EMDQ, NVH, UE, EEU, EEP, PQ, A&U, TIGHT, ENVI, WT, AERO, THM, DUR, CORR, SAF LOW = THM, DUR, VD, WT, CORR, SAF
Armenia - Yerevan	19-Nov-27	VF6 Legacy	LHD	Capital market with strong road-safety focus and mixed urban infrastructure.	Seasonal environment exists, but the strongest evidence points to safety and vehicle robustness needs rather than a single thermal extreme.	Safety burden remains material; suspension resilience, braking stability, and crash-avoidance behavior matter strongly.	Driver expectation should be assumed to favor control confidence, repairability after minor impact, and durable city operation.	HIGH = DUR, VD, SAF, ADAS MEDIUM = THM, EMDQ, WT, CORR, NVH, UE, EEU, EEP, PQ, A&U, TIGHT, ENVI, AERO LOW =
Georgia - Tbilisi	20-Aug-27	VF6 Legacy	LHD	Capital-city taxi market with mixed infrastructure quality and meaningful road-safety focus.	Seasonal climate exists, but the bigger issue in the sources is safety/road-use pattern rather than temperature extremes.	Road safety burden remains significant; robustness and repairability matter in mixed-infrastructure use.	Expect dense city maneuvering, minor-contact risk, and strong importance of braking confidence and repair turnaround.	HIGH = DUR, VD, SAF MEDIUM = THM, EMDQ, WT, CORR, NVH, UE, EEU, EEP, PQ, A&U, TIGHT, ENVI, AERO, ADAS LOW =



Comparison Table of GSM Markets – Central Asia

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Country	Launch date	Proposed vehicle from VDD	LHD / RHD	Geography / urban context	Climate & temperature condition	Road & infrastructure condition	Driving behavior & customer expectation	Attribute priority for market and risk
Uzbekistan - Tashkent	19-Nov-27	VF6 Legacy	LHD	Major urban center with road-network expansion because traffic loads are already high.	Strong seasonal swing is likely, but the sources emphasize traffic load and road-program expansion more than climate detail.	High traffic intensity, corridor overload, and mixed pavement quality suggest heavy wear exposure.	Driving pattern is likely high-utilization urban + corridor duty, with stop-go heat load and dependence on predictable operating economics.	HIGH = THM, DUR, VD, EMDQ, ADAS MEDIUM = WT, CORR, SAF, NVH, UE, EEU, EEP, PQ, A&U, TIGHT, ENVI, AERO LOW =
Kyrgyzstan - Bishkek	17-Sep-27	VF6 Legacy	LHD	One of the harshest operating markets in your list; city and surrounding network imply rough-duty usage.	Cold season exposure is material; source reporting on road works explicitly references the onset of cold weather.	Road quality is among the weakest in your launch set, and even repair quality is publicly questioned.	Driver behavior/expectation will favor a car that can survive impacts, keep alignment, and avoid squeaks/rattles under hard use.	HIGH = THM, DUR, VD, EMDQ, SAF, NVH MEDIUM = WT, CORR, UE, EEU, EEP, PQ, A&U, TIGHT, ENVI, AERO, ADAS LOW =





Comparison Table of GSM Markets – Asia-Pacific

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Country	Launch date	Proposed vehicle from VDD	LHD / RHD	Geography / urban context	Climate & temperature condition	Road & infrastructure condition	Driving behavior & customer expectation	Attribute priority for market and risk
Australia - Sydney	20-Aug-27	VF6N	RHD	Coastal developed-city taxi market, RHD, strong fleet-service expectations.	Humid subtropical; no severe winter, moderate summer heat, rainfall spread across the year.	Roads and public EV ecosystem are relatively favorable; launch risk is more operational than environmental.	Customers/fleets care about uptime, software stability, charging compatibility, and smooth taxi operation more than extreme-duty survivability.	HIGH = MEDIUM = THM, DUR, VD, EMDQ, SAF, NVH, UE, EEU, EEP, PQ, A&U, TIGHT, ENVI, WT, AERO, ADAS LOW = WT, CORR
Australia - Melbourne	3-Dec-27	VF6N	RHD	Developed-city taxi market, RHD, variable weather and event-driven urban demand.	Temperate maritime with mild winters, warm summers, and occasional heat waves.	Roads are generally manageable; weather variation is more important than road collapse.	Drivers will notice HVAC response, demist/defog behavior, brake/grip consistency in changeable weather, and software polish.	HIGH = MEDIUM = THM, DUR, VD, EMDQ, SAF, NVH, UE, EEU, EEP, PQ, A&U, TIGHT, ENVI, WT, AERO, ADAS LOW = WT, CORR
Hong Kong	TBU	Limo Green	RHD	Hyper-dense coastal city built around Hong Kong Island, Kowloon, and the New Territories, with very limited flat land, dense development, reclaimed urban zones, and hilly / mountainous terrain outside the main urban core. Hong Kong is one of the world's most vertically developed and transit-oriented cities.	Sub-tropical climate with hot, humid summers, frequent showers / thunderstorms, and afternoon temperatures often above 31°C from May to August; winter is milder but occasional cold fronts can bring urban temperatures below 10°C.	Road network is heavily used, with about 2,271 km of public roads, 24 road tunnels, major bridges, dense development, and undulating terrain that create maintenance and traffic challenges. Real-time government traffic systems and frequent traffic notices indicate sustained congestion, works, diversions, and high strategic-route loading. Hong Kong also has dense traffic, narrow hill roads, and extensive tunnel use	driving is challenging because of congestion, narrow lanes, steep terrain, tunnels, and complex interchanges. Customer expectation for an e-taxi is likely high because the Government is actively subsidising 3,000 e-taxis, planning more fast chargers, and expanding the EV roadmap. This means customers/operators will expect strong low-speed drivability, tight urban maneuverability, high uptime, fast charging turnaround, and premium cabin comfort in dense city use	HIGH = THM, VD, EMDQ, TIGHT, SAF, NVH, ADAS MEDIUM = DUR, CORR, UE, EEU, EEP, PQ, A&U, ENVI, WT, AERO LOW =

Thank You

